$Ax = b$

$AA^{-1}x = A^{-1}b$

$x = A^{-1}b$

$$\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}\underset{A}{\begin{bmatrix} \quad & \quad \\ \quad & \quad \end{bmatrix}} = I$$

$$\xleftarrow[\overset{\downarrow}{A^{-1}}]{E}$$

$A^{-1}A = I$

$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$ linearly dependent cols

So why does that stop us from finding Inverse?

$$\begin{tikzcd}
x_1 \arrow[r] & f(x) \arrow[r] & y \arrow[d, bend left] \\
x_2 \arrow[ur] & f^{-1}(x) \arrow[l] & \\
?\ x_1 \arrow[d] & & \\
?\ x_2 & &
\end{tikzcd}$$

2 inputs getting mapped to single value.

want non zero vector

$$\underset{A}{\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}}\underset{x}{\begin{bmatrix} x \\ y \end{bmatrix}} = \underset{0}{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}$$

$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} \rightarrow \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

2 diff vectors getting mapped to single output

A whole line will fall into 0.

For 3d → a plane.

Let's assume A has an inverse

$$Ax = 0$$

$$A^{-1}Ax = A^{-1}0$$

$$x = 0$$

$$\textcircled{A}A^{-1} = I$$

$$A^{-1}\textcircled{A} = I$$

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_{I}$$

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} a & b \\ c & d \end{bmatrix}}_{A^{-1}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} a \\ c \end{bmatrix}}_{x_1 = b_1} + \underbrace{\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} b \\ d \end{bmatrix}}_{x_2 = b_2}$$

$$Ax_1 = b_1$$

$$Ax_2 = b_2$$

$$\left[\begin{array}{cc|cc} \boxed{1} & 2 & 1 & 0 \\ 3 & 5 & 0 & 1 \end{array}\right] \xrightarrow[E_{21}]{} \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array}\right] \; \times(-1)$$

I augmented on right.

$$\xrightarrow{E} \left[\begin{array}{cc|cc} \textcircled{1} & 2 & 1 & 0 \\ 0 & \textcircled{1} & 3 & -1 \end{array}\right] \Rightarrow r_1 = r_1 - 2 \times r_2$$

$$= \left[\begin{array}{cc|cc} 1 & 0 & -5 & 2 \\ 0 & 1 & 3 & -1 \end{array}\right]$$

I

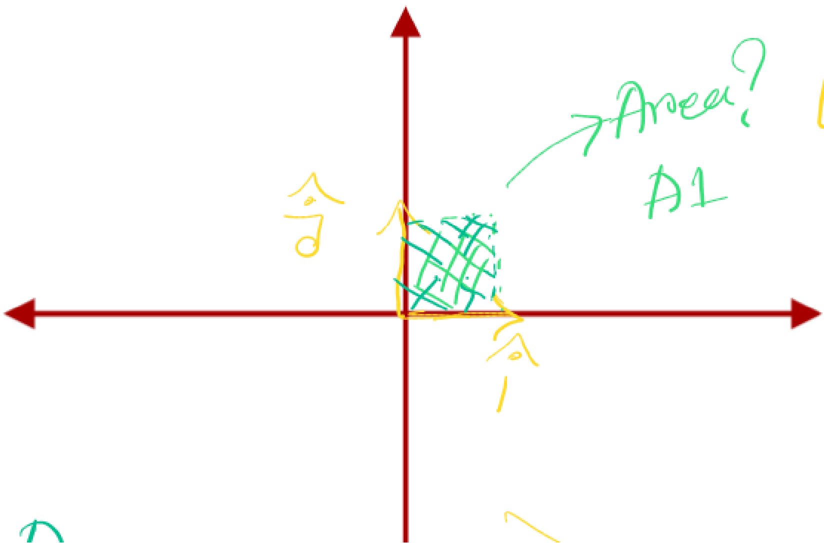
$$\underbrace{E}_{\downarrow \; A^{-1}} [A \mid I] = [I \mid ?] \quad ? \to \boxed{E} \to A'$$

LUd composition.

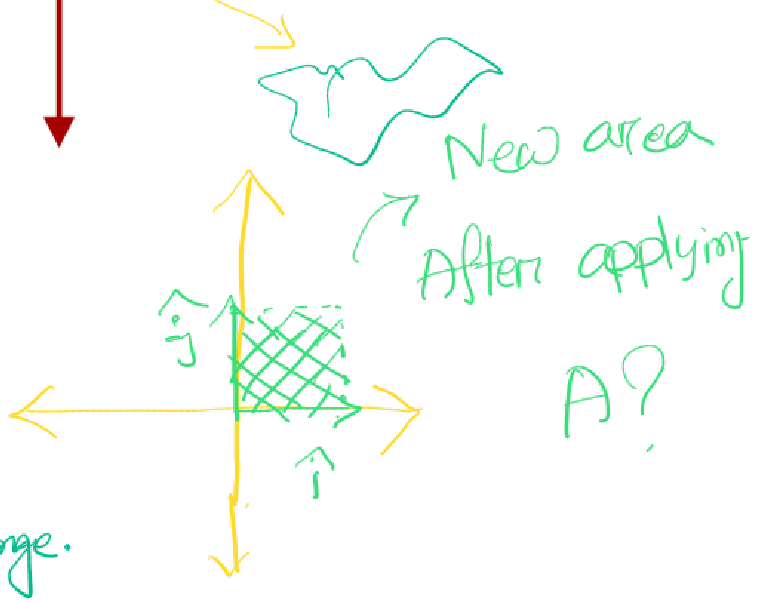
Determinants

$\to$ How much the area will b scaled by?

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$



$A_2 = 4 \cdot A_1$



$\hat{i}$, $\hat{j}$ flip : Det sign change.